BIOL-1 Practice Test 04

Exam 03 Preparation (Modules 12–15)

Instructions: This practice test covers material from **Module 12** (Darwin and Evolution) through **Module 15** (Population and Systems Ecology). Answer all questions to the best of your ability to prepare for Exam 03.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 12: Darwin and Evolution

1. In the logic of natural selection, which observation states that more offspring are produced than can usually survive to reproduce?
A) Differential reproductive success
B) Overproduction of offspring
C) Inheritance of acquired characters
D) Stabilizing selection only

2. A trait increases **evolutionary fitness** in a given environment if individuals with that trait tend to:
A) Look larger than their competitors
B) Contribute more alleles to the next generation than competitors do
C) Acclimate in one lifetime to match the parent phenotype
D) Survive long enough to be counted, even with zero offspring

3. Which statement is **most** consistent with **descent with modification**?
A) Each species was created in its final form and does not change
B) Lineages change over time; related species share ancestors with modification
C) Only fossils evolve, not living species

D) The strongest individual in a species can change the population's heritable gene pool in one lifetime, without reproduction

4. Homologous forelimb bones in a human, a cat, and a whale are used as evidence for evolution because they:
A) Show identical functions in all three species
B) Suggest a shared ancestry with structural change for different uses
C) Prove that whales evolved from modern humans
D) Disprove the existence of a fossil record

5. A population of moths in a clean forest is mostly light-colored; in a sooty industrial region the population becomes mostly dark. The clearest **selection** label for that shift, if the environment favors the dark extreme, is:
A) Stabilizing selection
B) Disruptive selection
C) Directional selection
D) The founder effect

6. Which option best states why **populations** evolve, not **individuals**?
A) Individuals can change their DNA by lifting weights
B) Evolution is a change in allele **frequencies in a population** across generations
C) One hungry bird cannot affect allele frequencies at all (bird effects are not evolution)
D) Natural selection has nothing to do with heritable variation

Module 13: How Populations Evolve

7. The definition of **microevolution** used in this course is:
A) A change in an individual's size during one lifetime
B) A **change in allele frequencies** in a population over time
C) The origin of a new phylum in one generation
D) A synonym for any mutation in a body cell only

8. A few finches from the mainland start a new island population. The island gene pool reflects only the alleles the founders brought. This is:
A) The bottleneck effect
B) The **founder** effect (genetic drift)

C) Stabilizing selection on beak size

D) Convergent evolution

9. A volcanic eruption kills 95% of a squirrel population at random, then survivors repopulate. The loss of many alleles from the pre-eruption pool is:

A) A founder effect in a new island colonization from another continent only

B) A **bottleneck** (genetic drift)

C) A proof that survivors were the "most fit" for every possible future environment

D) Interspecific competition in the absence of intraspecific variation

10. In **stabilizing** selection on a trait (for example, birth weight in some human data sets):

A) Both extreme phenotypes are favored

B) The **intermediate** phenotype is favored; extremes are selected against

C) The population's average always moves to one tail of the range

D) Allele frequencies must stay exactly at Hardy–Weinberg equilibrium forever

11. The Hardy–Weinberg equilibrium model is useful mainly because it:

A) Describes a real population that never evolves

B) Serves as a **mathematical baseline** to measure evolutionary change in real data

C) Replaces the biological species concept

D) Guarantees $p^2 + 2pq + q^2 = 1$ in every generation without assumptions

12. The movement of alleles into or out of a population through **migration and mating** is:

A) Non-random mating only

B) **Gene flow**

C) Genetic drift in an infinitely large population

D) A postzygotic barrier to reproduction

Module 14: Macroevolution

13. Under the **biological species concept**, two groups are different species if they:

A) Look different in a field guide, regardless of interbreeding

B) Are **reproductively isolated** in nature (they do not interbreed to produce viable, fertile offspring **between** the two groups, while within each group interbreeding can occur)

C) Have different number of cells

D) Live on different trees in the same forest, without any other test

14. Fireflies of two species use different **flash** patterns, so they rarely attempt mating. This isolating mechanism is best classified as:

A) Postzygotic reduced hybrid fertility

B) A **prezygotic behavioral** barrier

C) A mechanical postzygotic barrier

D) The bottleneck effect

15. A mule, the hybrid of a horse and a donkey, is typically sterile. This is an example of:

A) A prezygotic gametic barrier

B) A **postzygotic** barrier to gene flow (reduced hybrid fertility)

C) Stabilizing selection in horses

D) The founder effect

16. A new mountain range **physically splits** a population; over long time, the two lineages no longer interbreed. The classic geographic mode of speciation is:

A) Sympatric speciation in one meadow

B) **Allopatric** speciation

C) The bottleneck effect in both lineages on purpose

D) A requirement that drift never occurs

17. A plant can form a new, reproductively isolated **polyploid** lineage in a few generations, often in the same area as a parent type. This route is a frequent example of:

A) The bottleneck effect only

B) **Sympatric** speciation (e.g. via polyploidy)

C) A prezygotic postzygotic hybrid

D) Only macroevolution in animals, not plants

18. The statement “macroevolution is different chemistry from microevolution” is **false** in this course because:

A) No DNA is involved in speciation

B) The same **population**-level processes (selection, drift, flow, etc.) can accumulate to reproductive isolation and larger-scale diversity

C) All macroevolution happens in a single meiosis

D) The fossil record is irrelevant to modern biology

Module 15: Population and Systems Ecology

19. A population introduced to a new habitat with abundant resources for a time shows growth that **accelerates** (as long as the ceiling is not yet reached). That early phase is classic for:

A) A flat line at carrying capacity from year one only

B) An **exponential (J**-shaped) phase of growth

C) A carrying capacity of zero by definition

D) Only density-independent regulation, never K

20. A population levels off as resources become scarce, approaching a relatively stable size. That ceiling is:

A) A genetic bottleneck

B) **Carrying capacity (K)** in a logistic (S) model

C) A proof that 100% of energy moves to the next level

D) A synonym for the founder effect

21. A fungal disease that spreads more easily in **dense** host stands is an example of a factor often:

A) **Density-dependent** in its impact on the population

B) Only density-independent like wind

C) Unrelated to per-capita rates

D) Never biological

22. A severe freeze kills a **similar fraction** of a plant population across sparse and dense patches (the freeze does not "care" about crowding the way a contagion might). Treated as a **limiting** event in textbook classification, the freeze is **most** like:

A) Only density-dependent competition for light

B) A **density-independent** disturbance (still harmful to the population)

C) A trophic level

D) A biogeochemical cycle for nitrogen only

23. The **~10% rule** between trophic levels implies that, compared to eating **producers** directly, a diet of **higher** trophic levels (many steps) to deliver the same human calories will tend to need:

A) **More** total primary production at the base of the food web

B) Less primary production because carnivores are 100% efficient
C) The same amount of energy at every level, by definition of food chains
D) Decomposers to add energy from the sun to dead matter

24. In an ecosystem, **chemical elements** (e.g. carbon, nitrogen in nutrients) are **recycled**, while the energy fixed by **photosynthesis** is eventually:
A) Re-used indefinitely by the same photon inside every cell with no loss
B) **Lost** largely as **heat** along the way; not recycled the way a carbon atom can be
C) Only stored in rocks, not in life
D) Transferred 100% from producer to all consumers combined

25. Fungi and bacteria in soil return nutrients from dead material to **forms** plants can use again. In contrast to energy, this recycling pattern supports the idea that:
A) Energy cycles exactly like water in the stratosphere only
B) Decomposers are central to **nutrient** cycling for producers, while **energy** flows in one direction through the system
C) Producers do not use soil nutrients
D) Decomposers are always apex predators in every web

Part B: Fill in the Blank (10 questions)

Word Bank: Allopatric, allele frequencies, decomposers, density-dependent, gene flow, homologous, K, prezygotic, stabilizing selection, sympatric

26. A population's long-term, stable ceiling in a logistic (S) model is the carrying capacity, symbol _______________.

27. The formal focus of **microevolution** is a change in ___________________ over generations in a population.

28. A barrier that works **before** a zygote—such as different breeding times—is a ___________________ barrier to reproduction (category name).

29. A limiting factor (such as a contagious disease) whose impact on survival or reproduction can intensify with crowding is often ___________________.

30. The movement of alleles into or out of a population through migration and mating is called __________________.

31. The forelimb bone pattern shared by many tetrapods with different functions (human arm, bat wing, whale flipper) is __________________ structure (in comparative anatomy, from common ancestry with modification).

32. Fungi and many bacteria, which break down dead material and return elements to the soil, are __________________ in nutrient cycling (category name).

33. Modes of natural selection: when the **middle** of a trait's range is favored and the extremes are selected against, the pattern is often __________________.

34. A river splits a continuous population; the two sides evolve in isolation. That geographic mode of speciation is often __________________ speciation.

35. A new species arises **without** geographic separation, as in some plant **polyploidy** lineages, is a textbook example of __________________ speciation.

Part C: Short Answer (4 questions)

36. (Module 12) In two or three sentences, define **evolutionary fitness** as used in this course, and name **one** line of **evidence** for common descent (fossils, comparative anatomy, biogeography, or molecular) with a brief what-it-shows.

37. (Module 13) Distinguish **bottleneck** and **founder** effects. Then name the three **modes of natural selection** on a continuous trait (directional, stabilizing, disruptive) in one line each.

38. (Module 14) Give one **prezygotic** and one **postzygotic** example of a **reproductive barrier** (one sentence each). In one sentence, contrast **allopatric** and **sympatric** speciation.

39. (Module 15) Contrast **density-dependent** and **density-independent** limiting factors with one **example of each** (one sentence on each type). In two sentences, use the **~10%** energy rule to explain **why** food chains with many links support **fewer** apex consumers than a short chain from the same primary production.

End of Practice Test 04